Jacy grammar documentation: The book

Melanie Siegel

THE SLOW SCIENCE MANIFESTO

- Science needs time to think. Science needs time to read, and time to fail. Science does not always know what it might be at right now. Science develops unsteadily, with jerky moves and unpredictable leaps forward—at the same time, however, it creeps about on a very slow time scale, for which there must be room and to which justice must be done.

http://slow-science.org/

Jacy grammar history

- 1995-2000
  - Verbmobil: Machine translation of application-oriented spoken dialogues.

    http://verbmobil.dfki.de/
- 2001-2002
  - Co-operation with YY Technologies (CA, USA): Automatic email response (Co-operation with Stephan Oepen, Ulrich Callmeier, Monique Sugimoto, Atsuko Shimada, Dan Flickinger)
- 2002-2004
  - EU project DEEP THOUGHT: Hybrid and shallow methods for knowledge-intensive information extraction

Jacy grammar history

- 2004-2007
  - Hinoki: Automatic ontology extraction, treebanking and lexical type database (Co-operation with NTT Japan)
- 2006
  - Habilitation at University of Bielefeld: "JACY - A Grammar for Annotating Syntax, Semantics and Pragmatics of Written and Spoken Japanese for NLP Application Purposes"
    http://www.melaniesiegel.de/publications/jacy-documentation.pdf
- 2005-2007
  - Machine Translation Japanese-English, English-Japanese, Japanese-Norwegian and Korean-Japanese (Co-operation with Logon)
- 2006-2015
  - Work on the book, Emily, Francis an me (while doing a lot of other things)

As these are really long-term projects…

WHY DO WE PRODUCE HPSG GRAMMARS?

Jacy - The Book (Melanie Siegel, Emily Bender and Francis Bond)

Applications

- HPSG grammars for better NLP applications
  - more precise information on the semantics of language
  - information extraction, machine translation, email categorization, dialogue understanding, question answering
  - interfaces to other components (e.g., NER, POS tagging) for robustness
  - automatically producing annotated data (treebanks) that enables further research

Scientific insights

- HPSG grammars to better understand languages
  - analyzing the systematic and semantic differences between languages.
    - Does that tell anything about culture / the people?
    - What is the part that is common for some / all languages?
  - documentation of endangered languages

Scientific insights

- HPSG grammars to better understand linguistics
  - what is the connection between morphology, syntax, semantics, and pragmatics?
  - testing linguistic hypotheses:
    - validating theoretical claims against large amounts of data
    - show the relevance of such theoretical claims to language as spoken (or written) by native speakers

As this is a lot of work…

WHY DO WE DOCUMENT THE GRAMMARS?

Basis for new grammars

- the matrix
- grammars for languages with similar phenomena, like Korean
- finding similiar or different solutions for similar phenomena in other languages, being able to compare

HPSG in applications

- defining interfaces
  - for preprocessing, such as NER or POS tagging
  - for post-processing, such as zero pronoun resolution or translation
- understanding the output of the grammars

Documentation of our work – scientific reputation

- book: possibility to write up the whole story – not restricted to 5.000 words

WHAT SHOULD BE DOCUMENTED?

Book

1 Introduction

1.1 History of the Grammar

1.2 Theoretical Framework

1.3 Jacy among other HPSG Grammars of Japanese

1.4 Jacy and Multilingual Grammar Engineering

1.5 How this Book is Structured

1.6 Further Reading

Book

2 JACY in Different Application Domains

2.1 Appointment Scheduling in Machine Translation

2.2 Email in the Banking Domain

2.3 Hybrid Language Processing

2.4 Dictionary Denfition Sentences

2.5 Open Source Machine Translation

2.6 The Full Jacy System

2.7 Summary and Further Reading

Book

3 Basic Japanese Phrase Structure

- 3.1 Overview of HPSG
- 3.2 Core Phrase Structure Rules
  - 3.2.1 Head-Subject Phrase
  - 3.2.2 Head-Complement Phrases
  - 3.2.3 Head-Adjunct Phrase
  - 3.2.4 Coordinated Structures
  - 3.2.5 Head-Specifer Constructions
  - 3.2.6 Head-Marker Constructions
  - 3.2.7 Utterance Rules
- 3.3 Head-Initial Constructions
  - 3.3.1 The Position of Syntactic Heads in Japanese
  - 3.3.2 Head-Initial Modification
  - 3.3.3 Head-Initial Complementation
  - 3.3.4 Corpus Study
- 3.4 Semantic Representations
- 3.5 Summary and Further Reading

Book

4 Word Order and Subcategorization

- 4.1 Properties of Japanese arguments
- 4.2 Previous analyses
- 4.3 Jacy's approach
  - 4.3.1 The feature opt
  - 4.3.2 Required adjacency
  - 4.3.3 Zero pronominalization and obligatory overt realization
- 4.4 Summary and Further Reading

Book

5 Verbs and Adjectives

- 5.1 Verbal Subcategorization Types
  - 5.1.1 Intransitive Verbs
  - 5.1.2 Transitive Verbs
  - 5.1.3 Ditransitive Verbs
  - 5.1.4 Copula Verbs
  - 5.1.5 Adjective Subcategorization Types
- 5.2 Inflectional and Derivational Rules
  - 5.2.1 Stem Classes
  - 5.2.2 Ending Types
  - 5.2.3 Inflectional Rules
  - 5.2.4 Sample Derivation
  - 5.2.5 Stand-Alone Stems
  - 5.2.6 Derivational Rules
  - 5.2.7 Adjective Inflectional Types
- 5.3 Auxiliary Constructions
  - 5.3.1 Aspect Auxiliaries
  - 5.3.2 Content Auxiliaries
  - 5.3.3 Perspective Auxiliaries
  - 5.3.4 Auxiliaries and Honorifcation
  - 5.3.5 Contracted Auxiliaries
  - 5.3.6 Summary
- 5.4 Passive Constructions
  - 5.4.1 Simple Passives
  - 5.4.2 Adversative Passive
  - 5.4.3 Honorfication with Passive Morphology
  - 5.4.4 Potential with Passive Morphology
- 5.5 Causative
- 5.6 Summary and Further Reading

Book

6 Nominal Structures: Linking Syntax, Semantics and Pragmatics

- 6.1 Ordinary Nouns and Noun Phrases
- 6.2 Names and Named Entities
- 6.3 Pronouns
  - 6.3.1 Demonstrative Pronouns
  - 6.3.2 Locative Pronouns
  - 6.3.3 Personal Pronouns
  - 6.3.4 The Reflexive jibun
- 6.4 Nominalizations
- 6.5 Temporal Nouns
- 6.6 Noun Modification
  - 6.6.1 Noun Modification by Post-Positional Phrases
  - 6.6.2 Relative Clause Constructions
  - 6.6.3 Pre-Nominal Adjectives
- 6.7 Numeral Classifiers
  - 6.7.1 Semantic Representations
  - 6.7.2 The Analysis
- 6.8 Summary and Further Reading

Book

7 Particles

7.1 Co-occurrence of Particles

7.2 The Type Hierarchy of Japanese Particles

7.3 Case Particles case-p-lex

- 7.3.1 General Properties of Case Particles
- 7.3.2 The Case Particle ga
- 7.3.3 The Case Particle wo
- 7.3.4 The Case Particle ni
- 7.3.5 Complementizers
- 7.3.6 Other Case Particles

7.4 Particles with Semantic Content cont-p-lex

- 7.4.1 Genitive-Specifying no
- 7.4.2 Modifying Particles: mod-p-lex
- 7.4.3 Noun-Modifying Particle no nmod-p-lex
- 7.4.4 Verb-Modifying Particles vmod-p-lex
- 7.4.5 Noun Phrase Conjunctions conj-p-lex

7.5 Sentence Particles and Sentence Force sa-p-lex

7.6 Omitted Particles

7.7 Evaluation of Case and Modifying Particles

7.8 Summary and Further Reading

Book

8 Adverbs and Demonstratives

8.1 Adverb Morphology

8.2 Types of Adverbial Modification

8.3 Interrogatives

8.4 Demonstratives

8.5 Summary and Further Reading

Book

9 Honorifics

9.1 Honorific Forms in Japanese

- 9.1.1 Three Types of Honorifics
- 9.1.2 Interaction of Different Kinds of Honorification

9.2 Previous Approaches

9.3 Japanese Honorification in Jacy

9.4 Evaluation

9.5 Honorification and Machine Translation

9.6 Honorification in Other Languages

9.7 Summary

Book

10 Grammar Engineering

- 10.1 The Development Environment
- 10.2 The Grammar Development Cycle
- 10.3 Integrating Treebanking
  - 10.3.1 Selecting the Correct Parse
- 10.4 Embedded Documentation: The Lexical Type Database
  - 10.4.1 Linguistic Discussion
  - 10.4.2 Exemplification
  - 10.4.3 Implementation
  - 10.4.4 Building the Lexical Type Database
- 10.5 Automatic Error Detection and Correction
- 10.6 Summary and Further Reading

Book

11 The Current State of the Grammar

11.1 What is the Grammar Size? How Many Rules and Lexical Entries does it Contain?

11.2 What is the General Coverage on What Kinds of Data?

- 11.2.1 Evaluation Using Treebanks
- 11.2.2 Coverage Evaluation
- 11.2.3 Test Suites

11.3 How Far is the Grammar Flexible and Useful for Applications? Is it Domain-Adaptable and Can Be Used for Different Application Domains?

11.4 How Far Can the Grammar Be Used in Multilingual Applications?

11.5 Is the Output Precise and does it Correspond to Semantic Format and Content Restrictions?

11.6 Summary and Further Reading

12 Conclusion

Linguistic types



adv-p-lex-np (ltype)

Lexical Examples: 3 (3 out of 7: more)

lexid	Lemma	Surface	Frequency
ni-narg	に	に,	2,679
de-advp-np	で	で,	884
comma-advp	、	、,	103

Corpus Examples (3 out of 3,666: more)

110552: 彼女 は 実権 の ない 重役 に 祭り上げられ ました (parse)

123250: 私 は この 件 で は あなた に 賛成 できない (parse)

123250: 私 は この 件 で は あなた に 賛成 できない (parse)

Type Information

```
adv-p-lex-np := adv-p-lex &
 [ SYNSEM  [ LOCAL  [ CAT  [ VAL  [ COMPS  [ FIRST  [ LOCAL  [ CAT  [ HEAD    noun_head & [ FIN  + ] ] ] ] ] ] ] ] ] ].
```

Supertypes	Head Category	Valence	Content	Subtypes	Arity	head
adv-p-lex	adv-p_head	p_sat	prep-sem-type	adv-p-lex-np-nonexh		

Linguistic Type Database for Jacy_1301; By Chikara Hashimoto and Francis Bond; Maintained by Francis Bond <bond@ieee.org>

Jacy - The Book (Melanie Siegel, Emily Bender and Francis Bond)

Linguistic types



http://compling.hss.ntu.edu.sg/ltdb/Jacy_1301/index.html

Lexical entries

- we should have a possibility to document lexical entries in an HPSG
- lexical entry database?

Performance

- some ways to document grammar performance:
  - online demo of the grammar
  - online demo of applications that include the grammar
  - tree banks that are built with the grammar
  - annotated corpora

HOW TO DOCUMENT THE GRAMMARS?

Book

- scientific discussion
- collaborative writing for a scientific discussion
- hundreds of marginpars in .tex documents
- hours of discussions of those
- hundreds of mails

Collaborative document (e.g., WIKI)

- while writing the grammar
- especially for collaborative development
- to get direct feedback on design decisions

Automatic documentation

- lexical type interface
  - to make unterstanding of technical details clearer
- corpus annotation
  - to demonstrate the performance of the grammar
- demo interface
  - to demonstrate the performance of the grammar
  - Heart of Gold (only English now ☹)
  - http://compling.hss.ntu.edu.sg/demophin/jacy/

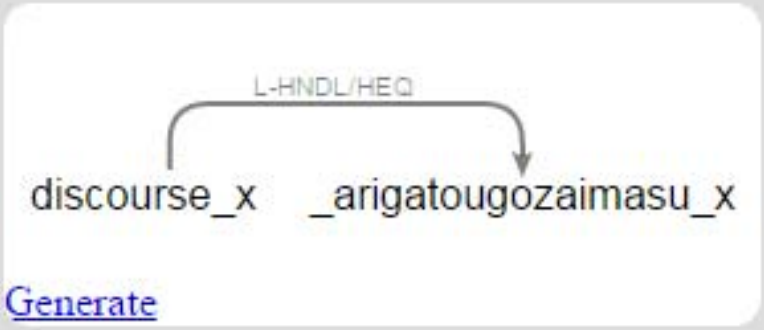
Next steps

- **finish the book, while we are here!**
- finish other books…
- implement more ways to document the grammar(s)
- set up more online demonstrators, in the spirit of the Heart of Gold
- implement and document more applications using our grammars

The Jacy Japanese Grammar

どうも　ありがとう　ございます　Parse

Parse results for "どうも　ありがとう　ございます"



1 readings, added 192 / 68 edges to chart (10 fully instantiated, 20 actives used, 9 passives used) RAM: 825k